

Data Source Manifest

This document provides an inventory of all data files used in the EMS Readiness Optimization project.

Raw Data Files

1. FDNY Firehouse Listing

Attribute	Value
Filename	FDNY_Firehouse_Listing_20260223.csv
Size	34 KB
Records	219 firehouses
Source	NYC Open Data
Download Date	February 23, 2026
Data Dictionary	FDNY_Firehouse_Listing_Data_Dictionary.xlsx
Description	Complete listing of FDNY firehouses across NYC, including location coordinates, facility names, and borough assignments

Key Fields:

- FacilityName: Name of the firehouse
- FacilityAddress: Street address
- Borough: NYC borough (Manhattan, Brooklyn, Queens, Bronx, Staten Island)
- Latitude/Longitude: Geographic coordinates

2. Motor Vehicle Collisions - Crashes

Attribute	Value
Filename	Motor_Vehicle_Collisions_- _Crashes_20260223.csv
Size	536 MB
Records	~2.24 million crash records
Source	NYC Open Data (NYPD)
Download Date	February 23, 2026
Data Dictionary	Mo- tor_Vehicle_Collisions_Data_Dictionary.xls x
Description	Comprehensive record of motor vehicle collisions in NYC reported by NYPD, including date, time, location, and injury information

Key Fields:

- CRASH DATE / CRASH TIME: Temporal information
- BOROUGH: NYC borough
- LATITUDE / LONGITUDE: Crash coordinates
- NUMBER OF PERSONS INJURED/KILLED: Casualty counts
- CONTRIBUTING FACTOR: Crash cause
- VEHICLE TYPE: Types of vehicles involved

3. Police Precincts

Attribute	Value
Filename	Police_Precincts_20260223.csv
Size	3.6 MB
Records	78 precincts
Source	NYC Open Data
Download Date	February 23, 2026
Data Dictionary	Police_Precincts_Data_Dictionary.xlsx
Description	Police precinct boundaries and information for NYC

Key Fields:

- Precinct: Precinct number
- the_geom: Polygon geometry for precinct boundaries

4. Manhattan Boundary

Attribute	Value
Filename	manhattan_boundary.geojson
Size	247 KB
Source	NYC Open Data / Derived
Description	GeoJSON polygon defining Manhattan borough boundary

5. CBD Boundary (MTA Congestion Relief Zone)

Attribute	Value
Filename	cbd_boundary.geojson
Size	101 KB
Source	MTA / NYC Open Data
Description	GeoJSON polygon defining the Central Business District (MTA Congestion Relief Zone) used for robustness analysis

6. NYC Borough Boundaries

Attribute	Value
Filename	nyc_borough_boundaries.geojson
Size	3.1 MB
Source	NYC Open Data
Description	GeoJSON polygons for all five NYC borough boundaries

Data Dictionary Files

Filename	Description
FDNY_Firehouse_Listing_Data_Dictionary.xlsx	Field definitions for firehouse listing
Motor_Vehicle_Collisions_Data_Dictionary.xlsx	Field definitions for crash data
Police_Precincts_Data_Dictionary.xlsx	Field definitions for precinct data

Data Lineage

Raw Data (data/raw/)



Interim Data (data/interim/)

- Manhattan-filtered crashes
- Manhattan firehouses
- Temporal aggregations



Processed Data (data/processed/)

- Demand matrices
- Network graphs
- Simulation inputs

Processed Data Files

crashes_manhattan.csv

Attribute	Value
Location	data/processed/
Size	99 MB
Records	416,434
Description	Crash records filtered to Manhattan boundary with <code>in_cbd</code> flag
Date Range	July 2012 - February 2026

firehouses_clean.csv

Attribute	Value
Location	data/processed/
Records	219 (all NYC)
Description	All firehouses with <code>in_manhattan</code> and <code>in_cbd</code> flags

firehouses_manhattan.csv

Attribute	Value
Location	data/processed/
Records	48 (27 in CBD)
Description	Manhattan firehouses only

precincts_manhattan.geojson

Attribute	Value
Location	data/processed/
Records	30 precincts
Description	Police precincts intersecting Manhattan

Data Quality Audit Findings

Audit Date: March 12, 2026

Crash Data Quality Issues

Issue	Count	Percentage	Impact
Missing coordinates	~448,804	20.0%	Cannot be geocoded; excluded from spatial analysis
Missing borough	~134,641	6.0%	Borough field unreliable; use spatial filtering instead
Invalid coordinates	~0.1%	<0.1%	Outside NYC bounds; excluded

Geographic Validation

Boundary	Status	Notes
Manhattan	✓ Valid	636.63 sq km; CRS EPSG: 4326
CBD	✓ Valid	259.58 sq km; 100% within Manhattan
Precincts	✓ Valid	All 78 geometries valid

Firehouse Data

- All 219 firehouses have valid NYC coordinates
- 48 firehouses in Manhattan (27 in CBD)
- No missing data

Borough Distribution (from raw crash data)

Borough	Crashes	Percentage
Brooklyn	~627,000	28.0%
Queens	~516,000	23.0%
Bronx	~294,000	13.1%
Manhattan	~278,000	12.4%
Staten Island	~79,000	3.5%
Missing	~450,000	20.0%

Note: Spatial filtering to Manhattan boundary yields 416,434 crashes with valid coordinates.

Update Log

Date	Action	Notes
2026-03-12	Initial data acquisition	All raw data files downloaded from NYC Open Data
2026-03-12	Data audit complete	Validated all datasets; created Manhattan-filtered processed files